

From Exponentials to Logarithms

Friday, October 2, 2026
Quiz 6 (Section 4.3): Wednesday, October 7

Name _____ Fill in each example as it is worked on the whiteboard.

Why these topics today

Class ALEKS progress: Section 4.2 is at 56% and Section 4.3 is at 18%. The lowest skills are continuous interest, graphing b^x with its asymptote, converting natural logarithms, and solving exponential equations with logarithms. Today targets exactly those skills.

Warm-up

1. Evaluate 2^{-3} and $(\frac{1}{2})^{-2}$.

2. Evaluate $5000(1.04)^3$ to the nearest cent.

3. Use a calculator: $e^{0.5} \approx$ _____

Part 1 · Exponential functions (Section 4.2)

An exponential function has the form $f(x) = b^x$ where $b > 0$ and $b \neq$ _____.

y-intercept: _____

Domain: _____

Horizontal asymptote: _____

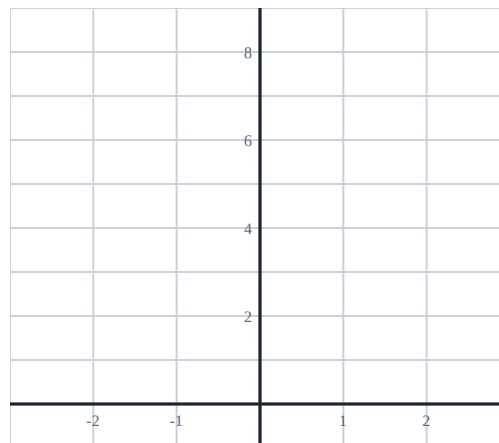
Range: _____

If $b > 1$ the graph shows _____; if $0 < b < 1$ the graph shows _____.

Whiteboard Example 1

Complete the table, plot both functions on the grid, and draw the asymptote as a dashed line: $f(x) = 2^x$ and $g(x) = (\frac{1}{3})^x$.

x	-2	-1	0	1	2
$f(x) = 2^x$					
$g(x) = (\frac{1}{3})^x$					



Which function is growth? _____ Which is decay? _____

Your turn

For $h(x) = 3^x$, find $h(-2)$, $h(0)$, and $h(2)$. State the asymptote and whether h shows growth or decay.

Three interest models

P = principal, r = annual rate as a decimal, t = years, n = compounding periods per year.

Model	Formula	On Test 2?
Simple interest		
Compound interest		
Continuous compounding		

Common values of n : annually _____, quarterly _____, monthly _____, daily _____.

Whiteboard Example 2

\$2000 is invested at 5% for 6 years. Find the balance with (a) simple interest, (b) interest compounded quarterly, and (c) interest compounded continuously.

Your turn

\$5000 is invested at 4% for 10 years. Find the balance if interest is compounded monthly, and if it is compounded continuously.

Growth by r per period: multiply by _____ Decay by r per period: multiply by _____

Whiteboard Example 3

(a) A \$24,000 car loses 15% of its value each year. Find its value after 4 years. (b) $P(t) = 18000(1.023)^t$ gives a town's population t years after 2020. Estimate the population in 2030.

Your turn

A 60 mg dose of medicine decreases by 12% each hour. How much remains after 5 hours?

Part 2 · Logarithms (Section 4.3)

Key idea: a logarithm is an _____. $\log_b x$ answers the question “ b to what power gives x ?”

$$\log_b x = y \iff \underline{\hspace{2cm}}$$

$\log x$ means base _____ (common log) $\ln x$ means base _____ (natural log)

Restrictions: the base satisfies $b > 0$, $b \neq 1$, and the input satisfies x _____.

Whiteboard Example 4

Rewrite each equation in the other form.

Given	Other form
$\log_2 32 = 5$	
$10^{-2} = 0.01$	
$\ln x = 3$	
$e^{0.4} = y$	

Your turn

(a) $\log_6 36 = 2$

(b) $10^{1.5} = x$

(c) $\ln 8 = x$

(d) $e^{-1} = y$

Whiteboard Example 5

Evaluate without a calculator. Ask: “the base to what power gives the input?”

$\log_3 81 = \underline{\hspace{2cm}}$

$\log_5 \frac{1}{25} = \underline{\hspace{2cm}}$

$\log 1000 = \underline{\hspace{2cm}}$

$\ln e^7 = \underline{\hspace{2cm}}$

$\log_4 2 = \underline{\hspace{2cm}}$

$\log_9 1 = \underline{\hspace{2cm}}$

Why are $\log_2(-8)$ and $\ln 0$ undefined? _____

Your turn

$\log_2 64 = \underline{\hspace{2cm}}$

$\log 0.001 = \underline{\hspace{2cm}}$

$\log_7 \frac{1}{7} = \underline{\hspace{2cm}}$

$\ln \sqrt{e} = \underline{\hspace{2cm}}$

$\log_2 1 = \underline{\hspace{2cm}}$

$\log_8 2 = \underline{\hspace{2cm}}$

Whiteboard Example 7

Change of base: $\log_b a = \frac{\ln a}{\ln b}$ (or $\frac{\log a}{\log b}$). Round to four decimal places.

$\log 250 \approx \underline{\hspace{2cm}}$

$\ln 0.3 \approx \underline{\hspace{2cm}}$

$\log_2 20 \approx \underline{\hspace{2cm}}$

Reasonableness check for $\log_2 20$: $2^4 = \underline{\hspace{2cm}}$ and $2^5 = \underline{\hspace{2cm}}$, so the answer lies between 4 and 5. **Your turn:**

$\log_3 50 \approx \underline{\hspace{2cm}}$

Whiteboard Example 6

Solve. First convert to exponential form.

(a) $\log_x 49 = 2$

(b) $\log_2 x = -3$

(c) $\ln x = 2.5$

(d) $\log_3(2x - 1) = 2$

Your turn

(a) $\log_x 81 = 4$

(b) $\log_5 x = 3$

(c) $\ln x = -1.2$

(d) $\log(x + 4) = 2$

Whiteboard Example 8

Solve. Isolate the exponential expression first, then use logarithms. Round to four decimal places.

(a) $5^x = 42$

(b) $3e^{2x} = 45$

(c) When does \$2000 at 5% compounded continuously reach \$4000?

Your turn

(a) $4^x = 30$

(b) $6e^{0.3x} = 90$

Quiz 6 readiness check (no notes · mark each: ✓ secure · ~ correctable · ✗ not yet)1. \$3000 at 6% compounded continuously for 5 years.
Find A . ✓ ~ ✗2. Write $\ln y = 1.8$ in exponential form. ✓ ~ ✗3. Evaluate $\log_2 \frac{1}{16}$. ✓ ~ ✗4. Solve $2^x = 11$ (four decimal places). ✓ ~ ✗**My next ALEKS topic:** _____